Seller Cheat Sheet for IBM i

Core platform terms and concepts at-a-glance

WHAT IS IBM i?

IBM i is an integrated, secure, highly available operating system for IBM Power Systems. Formerly OS/400 (since 1988), it runs RPG, CL, COBOL, Java, Python, SQL, and Node.js on one platform with near-zero downtime. Over 6,000 companies run mission-critical workloads on it across banking, insurance, manufacturing, retail, distribution and government. Includes security, database, job and memory management and isolation, Rest API connectivity, all as part of the operating system.

FILE SYSTEMS: QSYS & IFS

QSYS.LIB — Native library structure. Everything is an Object stored in a Library (Library = folder, Object = typed file). Think of a Library as a folder, and Objects as the typed files inside it. Object types include *PGM (programs), *FILE (data), *CMD (commands), *SRVPGM (shared code), and more.

IFS — The IFS provides a UNIX-style hierarchical directory tree (e.g., /home/user/reports.pdf) alongside QSYS. It supports standard stream files just like Windows or Linux. Use cases include storing Java .class files, PDF reports, web content, XML/JSON data, and open-source application code. IFS enables IBM i to coexist with modern web and open-source toolchains.

Library — A named container in QSYS.LIB that holds objects.

Library List — The ordered search path of libraries a job uses to find objects. Like PATH.

Object — The fundamental unit; every resource has a type (*PGM, *FILE, ...). Objects in QSYS are hard typed, you cant change the extension to something else (makes IBM i immune from some security attacks).

QSYS — The system library of IBM-supplied objects. Never modified by users.

QTEMP — A per-job temporary library; objects are auto-deleted when the job ends.

IFS Root (/) — The top of the IFS tree; standard path syntax like /home/myapp/config.json.

DATABASE: Db2 FOR i

Physical File (PF) — The table that stores data on disk. Created with DDS or SQL CREATE TABLE. Lives in a library.

Logical File (LF) — A view or index over one or more PFs; stores no data. Like a SQL VIEW or INDEX.

File Member — A subset of data within a PF. Legacy concept; modern practice is one member per file.

Db2 for i — The relational database built into the OS. Supports ANSI SQL, stored procedures, triggers, and UDFs.

SQL Table — The modern way to create data with SQL. A SQL table and a DDS physical file are two ways to make the same *FILE.

SQL View — Created with CREATE VIEW; the SQL equivalent of a Logical File. Preferred for new development.

Journal (JRNL) — Tracks every change to a file; enables recovery, commitment control (transactions), and replication.

PROGRAMMING LANGUAGES

RPG — The signature IBM i business language. Evolved to ILE RPG (free-form); modern RPG reads like any structured language, with procedures, data structures, and embedded SQL (no external data connector needed, access to the data is handled though the OS).

CL (Control Language) — Scripting and automation. Issues commands, controls job flow, calls programs. Like a powerful shell script.

COBOL / Java / Open-Source — ILE COBOL (common in finance), embedded SQL, native Java, plus Python, Node.js, PHP, Ruby, and Perl via yum/dnf.

OPM RPG — Original Program Model (RPG II/III/400). Fixed-column legacy; still runs unchanged.

ILE RPG — RPG IV. Free-form syntax, procedures, data types, and SQL integration.

Free-Form — Modern RPG with no column restrictions; looks like C or JavaScript. Preferred for new work.

Fixed-Form — Classic positional-column RPG (C-, D-, F-specs). Legacy but still widely used.

CRTBNDRPG — Compiles a single-source RPG program straight into a *PGM object. The simplest compile path.

CRTRPGMOD — Compiles RPG into a *MODULE object; must then be bound with CRTPGM or CRTSRVPGM.

ILE: INTEGRATED LANGUAGE ENVIRONMENT

ILE — The modern program model. Binds programs in different languages (RPG, COBOL, C, CL) into one executable.

Module (*MODULE) — The compiled output of one source member; not directly callable.

Program (*PGM) — A runnable object created by binding one or more modules.

Service Program (*SRVPGM) — A shared library of procedures. Like a DLL on Windows or a .so on Linux.

Binding Directory — A list of service programs and modules to include automatically at bind time.

Activation Group — An isolated runtime within a job; controls object lifecycle and commitment-control scope.

DDS: DATA DESCRIPTION SPECIFICATIONS

DDS — A declarative language that defines file and screen layouts outside application programs. A schema language specific to IBM i.

DDS File Types — PF (database table), LF (view/index), DSPF (5250 screen), PRTF (printed report), ICF (system-to-system comms).

5250 Green Screen — The classic character-based terminal UI, defined by Display Files. Still valued for its reliability.

Subfile — A DDS construct for scrollable lists on a 5250 screen. Core to many IBM i UIs.
Externally Described — A program reads its file layout from a DDS/SQL object; definition changes propagate automatically.

Db2 FOR i SQL SERVICES

IBM i ships hundreds of built-in SQL views and functions (under QSYS2 and SYSTOOLS) so admins and developers query system info with standard SQL — no special utilities needed.

QSYS2 Schema — IBM-supplied SQL views and functions exposing system info: jobs, objects, message queues, journals.
QSYS2.JOBINFO — Active job details (status, user, CPU, subsystem). A programmatic WRKACTJOB.
QSYS2.OBJECT_STATISTICS — Object details (type, size, last used). Finds dormant or oversized objects.
QSYS2.SYSROUTINES — A catalog of all stored procedures, UDFs, and triggers.
SYSTOOLS Schema — Helper procedures for common tasks: spreadsheets, email, base64 encoding.
SYSTOOLS.GENERATE_SPREADSHEET — Runs a SQL query and writes the results to an IFS Excel (.xlsx) file.
ACS Run SQL Scripts — The free ACS tool to write, run, and save SQL scripts against IBM i.

DEVELOPMENT TOOLS

Code for IBM i — A free VS Code extension that turns VS Code into a full IBM i IDE: browse libraries, edit and compile source, view job logs, manage IFS.
VS Code — With Code for IBM i, connects over SSH with syntax highlighting, IntelliSense, and compile actions for RPG, CL, COBOL, DDS.
RDi (Rational Developer for i) — IBM's Eclipse-based IDE with advanced debugging, refactoring, and the LPEX editor.
ACS (Access Client Solutions) — Free IBM tool: 5250 emulator, SQL runner, schema browser, and file transfer. Every admin should have it.
IBM Navigator for i — A modern web GUI to manage jobs, users, disk, messages, performance, and system settings.
SEU — Legacy green-screen source editor. Replaced by Code for IBM i / RDi.
PDM — Legacy tool to list and act on source members. Being replaced by modern IDEs.
PASE for i — An AIX-compatible runtime; runs bash, Python, Node.js, and git natively on the server.
Open-Source (yum/dnf) — A package manager that installs Python, Node.js, git, nginx, and hundreds of packages onto IBM i.

JOBS & WORK MANAGEMENT

Job — The basic unit of work; every program runs in a job. Interactive, batch, or autostart.
Job Queue (JOBQ) — Where batch jobs wait to run; multiple queues can feed one subsystem by priority.
Subsystem — A runtime environment that controls how jobs run, isolation and memory management. E.g., QBATCH, QINTER, QSYSWRK.
Job Log — A record of every message and event in a job. Essential for debugging.
Spool File — Printed output held in a queue; can be viewed, emailed, or converted to PDF.
Message Queue (MSGQ) — An inbox for messages; the IBM i inter-process communication model.
WRKACTJOB — Work with Active Jobs. The IBM i equivalent of top on Linux.